Abdelrahman Khalifa

Systems Software Engineer — Rust · C · Linux · GSoC 2026 (Linux Foundation)

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SUMMARY

Systems software engineer working in Rust, C and modern C++ on Linux. Completed Google Summer of Code 2026 with The Linux Foundation (OpenPrinting), building the COSMIC desktop's printer stack and landing 7 PRs upstream in cups-rs and libcups. Comfortable with FFI, IPC, async code and maintainer code review.

OPEN SOURCE

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | May 2026 – Oct 2026

COSMIC printer setup tool in Rust; passed the final evaluation | github.com/Abd002/GSOC-2026 | github.com/Abd002/cosmic-printers

- Designed a printer-management stack as 5 Rust crates: `printers-core` (types, grouping queues into physical devices), `-client` (async varlink client), `-server` (CUPS, IPP, DNS-SD and Printer Application backend), `-ui` (iced/libcosmic) and `-app`.
- Exposed the backend over varlink IPC from `cosmic-settings-daemon`, so the C FFI code never loads in the Settings GUI process; the standalone app falls back to running the server in-process.
- Built a Printers page and an Add Printer flow for COSMIC Settings: live updates, print queue control, supply levels, and legacy printers set up through Printer Applications (PAPPL) instead of PPD drivers.
- Tested on real HP printers and Printer Applications (Ghostscript, Gutenprint, HPLIP); CI checks formatting, Clippy and tests on several architectures and runs CUPS, Avahi and `ippeveprinter` to test discovery.

Upstream contributions — OpenPrinting cups-rs (Rust) and libcups (C), COSMIC | 2026

- cups-rs: ported to libcups3 with CUPS 2 and CUPS 3 builds and CI; added IPP management operations, Printer Application support, connections by IP/port, `lpoptions` support and FFI bindings for the libcups3 DNS-SD API (PRs #18–#22, merged).
- libcups: fixed crashes in `cupsDNSSDDelete()` and a leaked client in the Avahi DNS-SD code (#164, #165, merged).
- COSMIC (in review): Printers page (cosmic-settings #2182) and varlink printers server (cosmic-settings-daemon #188).

Contributor — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules (#86977, merged).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553, merged).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214, merged).

EXPERIENCE

Embedded Software Engineer — EKSON | Aug 2025 – Present

Sole software engineer for 3-DoF and 6-DoF pneumatic motion simulators

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Built the PyQt5 desktop control app, cutting CPU usage and latency by **~40%** (measured) by moving serial I/O off the UI thread; added remote control and firmware updates through a Node.js/MongoDB backend.

Embedded Linux Systems Intern — Orange Digital Center (ODC) | Dec 2023 – Jan 2024

- Built Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing.

TECHNICAL SKILLS

Languages: Rust, C, C++ (modern), Python, Bash, JavaScript (Node.js)

Systems & Linux: Linux (Ubuntu/Debian), operating systems, FFI and bindings, IPC (varlink, D-Bus; zbus, zlink), concurrency (async/event-driven, multithreading), memory management, networking (TCP/UDP), CUPS/IPP, DNS-SD/mDNS (Avahi), kernel modules and device drivers

Tooling: Cargo, Clippy, rustfmt, Git/GitHub, GitHub Actions CI, CMake, Make, Yocto, Buildroot, QEMU, testing, debugging, performance optimization, code review

Desktop & embedded: iced, libcosmic, PyQt5; Zephyr, FreeRTOS, ARM Cortex-M (STM32), ESP32

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024 (Egyptian Collegiate Programming Contest).
- IEEE student branch Treasurer: managed a budget of more than $10k.

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | 2020 – 2025

Very Good (84.6%) with Honors. Graduation project: AUTOSAR-based V2X vehicle on STM32; I built the firmware update path and a dual-bank bootloader with rollback.